

POTCHEFSTROOM, South Africa

WMO: 683500

Lat: 26.7358S

Lon: 27.0756E

Elev: 1351

StdP: 86.11

Time Zone: 2.00 (E02)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.8	-0.2	-18.3	0.9	8.5	-14.4	1.3	12.6	8.8	21.1	7.8	19.5	1.2	20	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	34.2	17.1	32.8	17.1	31.6	17.1	21.6	27.0	20.7	27.0	20.1	26.8	3.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	17.5	23.6	19.0	16.3	22.6	18.2	15.5	22.1	70.0	26.8	66.7	27.2	64.3	26.7	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.3	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	23.2	22.3	21.2	17.7	14.1	11.1	10.8	14.0	17.8	20.6	21.6	22.7
	DBStd	5.20	2.18	2.04	2.33	2.63	2.52	2.76	3.04	3.69	3.54	3.28	2.70	2.44
	HDD10.0	61	0	0	0	1	4	19	27	9	1	0	0	0
	HDD18.3	847	1	1	5	42	133	218	235	139	51	16	5	2
	CDD10.0	3002	408	346	346	231	130	51	51	133	236	330	347	394
	CDD18.3	746	150	113	92	22	1	0	0	5	36	87	102	138
	CDH23.3	7950	1383	961	849	307	60	10	12	184	681	1101	1098	1304
(13)	CDH26.7	2968	577	353	287	51	4	0	0	28	228	457	442	542
(14)	Wind	WSAvg	2.6	2.6	2.4	2.1	2.0	2.1	2.2	3.0	3.2	3.3	3.3	3.2
(15) Precipitation	PrecAvg	607	106	88	70	54	15	6	3	9	19	51	81	96
	PrecMax	1074	254	239	209	160	110	42	22	82	118	143	166	216
	PrecMin	223	0	8	13	1	0	0	0	0	0	2	7	45
	PrecStd	165	53	54	39	42	23	9	6	16	26	35	43	40
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	33.7	33.5	30.4	27.4	24.9	25.1	29.3	32.4	35.2	35.4	35.9
		MCWB	18.0	17.3	17.8	16.7	14.7	12.4	11.4	12.7	14.1	16.1	16.6	17.0
	2%	DB	33.5	32.0	31.5	28.2	25.0	22.5	23.0	27.2	31.0	32.9	33.2	33.2
		MCWB	18.4	17.4	17.8	16.1	13.7	11.2	10.3	11.9	14.2	15.7	16.3	17.6
	5%	DB	31.8	30.6	29.8	26.7	23.7	21.0	21.2	25.4	29.5	31.1	31.2	31.6
		MCWB	18.8	18.1	17.8	15.6	13.2	10.5	9.9	11.5	13.7	15.5	16.6	17.7
	10%	DB	30.2	29.0	28.2	25.2	22.3	19.5	19.6	23.3	27.7	29.4	29.4	29.8
		MCWB	18.6	18.3	17.8	15.6	12.8	9.9	9.5	10.9	13.2	15.5	16.8	18.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	21.8	21.2	19.2	16.8	14.7	13.7	15.8	17.4	23.9	23.0	22.5
		MCDB	29.6	27.2	27.2	24.6	22.1	20.6	21.0	22.7	27.2	24.0	26.4	26.9
	2%	WB	21.0	20.8	20.3	18.2	15.3	13.1	12.2	13.9	16.2	19.7	20.6	21.0
		MCDB	28.7	27.0	27.1	24.4	22.4	19.4	19.6	23.1	26.4	26.4	25.8	26.4
	5%	WB	20.2	20.2	19.6	17.4	14.4	11.8	11.1	12.9	15.1	18.1	19.4	20.1
		MCDB	27.7	26.8	26.5	23.9	21.7	18.4	19.3	22.4	25.1	26.0	25.7	26.5
	10%	WB	19.7	19.7	18.9	16.6	13.5	10.7	10.0	11.9	14.1	17.0	18.5	19.3
		MCDB	26.9	26.1	25.5	22.9	20.4	17.9	18.4	21.9	24.6	25.1	25.5	26.4
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	12.6	13.1	14.2	16.5	17.4	17.6	18.0	17.6	15.4	14.1	13.1
		MCDBR	15.9	15.8	16.1	16.6	18.6	18.9	19.0	19.4	19.9	18.2	17.2	16.2
		MCWBR	5.3	5.6	5.5	6.5	8.3	9.4	9.4	8.9	8.1	6.9	5.7	5.6
	5% WB	MCDBR	12.6	11.9	12.4	13.1	15.0	16.0	16.4	16.7	16.3	14.0	12.3	12.3
		MCWBR	5.0	4.8	5.2	6.2	7.8	9.1	8.9	8.3	6.9	7.1	5.8	5.2
(40) Clear-Sky Solar Irradiance	taub		0.382	0.387	0.372	0.326	0.290	0.274	0.274	0.328	0.371	0.400	0.354	0.364
	taud		2.341	2.329	2.349	2.447	2.521	2.533	2.510	2.354	2.244	2.204	2.380	2.382
	Ebn at Noon		960	940	923	922	918	914	926	905	909	916	982	980
	Edh at Noon		135	133	124	102	86	80	85	110	135	150	129	130
(44) All-Sky Solar Radiation	RadAvg		6.84	6.34	5.64	4.78	4.30	3.88	4.23	5.04	6.09	6.57	7.00	7.10
	RadStd		0.71	0.57	0.47	0.39	0.20	0.17	0.12	0.25	0.26	0.44	0.55	0.41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.66	+0.75	N/A	N/A	N/A	N/A	N/A	-181	+234	+87

Nomenclature: See separate page